

SESA 6067

Spacecraft Orbital Mechanics

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1. Chapter 1: Orbital Maneuvers

1.1. Maneuvers

A maneuver is defined as **A controlled change in the velocity (magnitude and/or direction) of a spacecraft to alter its orbit**. Each mission contains multiple maneuvers and those maneuvers can be made of the following building blocks:

- Change orbit shape.
- Change orbit orientation (plane).
- Change of orbit phasing (anomaly).

There can be multiple reasons for maneuvers within a mission, including orbital transfers, maintenance and disposal. The goal of mission design is to **minimize ΔV within mission constraints**, these constraints being:

- Transfer time.
- Forbidden regions (e.g. radiation belts).
- Pointing requirements (panels, instruments).

1.2. Types of Maneuvers

This module will cover various types of coplanar transfers and plane change maneuvers, these are:

- Coplanar transfers
 - (Bi-elliptic) Hohmann transfer
 - Phasing maneuvers
 - Non-Hohmann transfers
 - Apse line rotation
- Plane change maneuvers
 - Inclination change
 - RAAN change

1.3. Impulsive Maneuver Assumption

For this module, each maneuver is assumed to be impulsive (kick maneuver). This assumes that the burn time is negligible in comparison to the orbital time. It is important to note that **ΔV cost of a mission is the sum of all maneuvers**. Further note that for this topic **ΔV is a vector and so must be added correctly**.

1.4. Hohmann Transfers

A Hohmann transfer is a form of co-planar maneuver that can be used to raise or lower an orbit. These types of maneuvers are typically used during launch to increase the height of an orbit to the desired altitude.

1.4.1. Circular Hohmann Transfer

A circular Hohmann transfer is used to move between two circular orbits via one intermediate elliptical orbit. The derivation for the orbital velocities of each orbit can be done via the angular momenta of each orbit. To start with a diagram of a circular Hohmann transfer is shown in **Figure 1**.

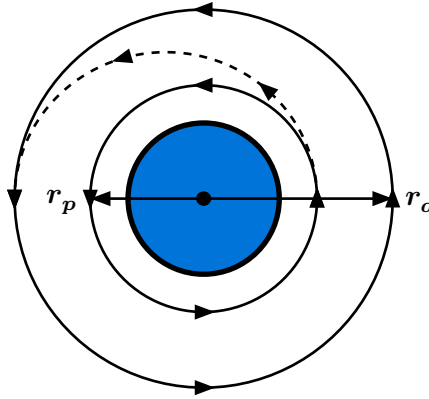


Figure 1: Schematic diagram of a circular Hohmann Transfer.

In order to calculate the Δv required for the two burns in this maneuver the polar orbit equation can be used, shown in **Eq. 1**, note that at perigee for the elliptical transfer orbit $\theta = 0^\circ$

$$r = \frac{p}{1 + e \cos(\theta)} = \frac{a(1 - e^2)}{1 + e \cos(\theta)} = \frac{h^2/\mu}{1 + e \cos(\theta)} \rightarrow r_p = \frac{h^2/\mu}{1 + e} \quad (1)$$

The expression in **Eq. 1** can be combined with the eccentricity equation shown in **Eq. 2**.

$$e = \frac{r_a - r_p}{r_a + r_p} \quad (2)$$

Combining both expressions **Eq. 1** and **Eq. 2** yield the following expression shown in **Eq. 3**.

$$r_p = \frac{a(1 - e^2)}{1 + e} \quad e = \frac{r_a - r_p}{r_a + r_p} \rightarrow h = \sqrt{2\mu \frac{r_a r_p}{r_a + r_p}} \quad (3)$$

Where h is the specific angular momentum ($m^2 s^{-1}$) of the orbit and r_p and r_a are the radius at perigee and apogee shown in **Figure 1**. Now consider the general expression for specific angular momentum shown in **Eq. 4**.

$$h = \vec{r} \times \vec{v} = rv \sin(\phi) \quad (4)$$

Where $\vec{r}$ and $\vec{v}$ are the position and velocity vectors and the angle ϕ is the angle between these two vectors. Importantly, $\phi = 0^\circ$ at the apogee and perigee points in the elliptical

cal transfer orbit. Combining **Eq. 3** and **Eq. 4** the velocities at r_p and r_a can be written in the form shown in **Eq. 5**.

$$v_a = \frac{h}{r_a} \quad v_p = \frac{h}{r_p} \quad \text{Where } h = \sqrt{2\mu \frac{r_a r_p}{r_a + r_p}} \quad (5)$$

Note that if $r_p = r_a$, meaning the transfer orbit is circular, then the expression simplifies to the following known expressions for circular orbits, shown in **Eq. 6**.

$$\text{If } r_a = r_p = r \quad \text{Then} \quad h = \sqrt{\mu r} \quad v = \sqrt{\frac{\mu}{r}} \quad (6)$$

1.4.2. Elliptical Hohmann Transfer

The derivation on the previous page is **not specific to circular orbits** as all of the equations used are general orbital equations. For this reason the equations can also be used for **transferring between two elliptical orbits** given they are coaxial (the Apse line of both orbits is the same).

However, there is a caveat in that it is not initially clear which transfer orbit is the most efficient. This is exacerbated if the two orbits are not in the same direction. This scenario would require calculating the deltaV of all 4 possible orbits (2 transfer orbits in each direction). The two possible orbits are shown in **Figure 2**.

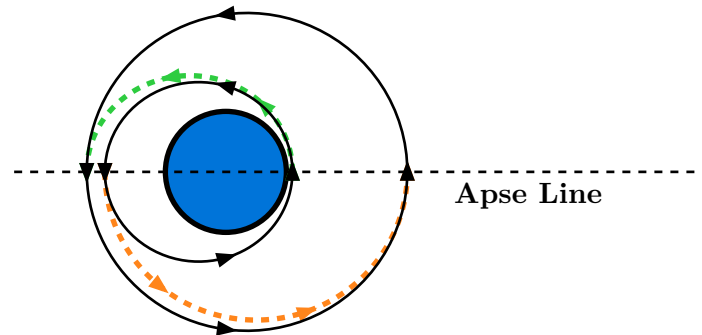


Figure 2: Schematic diagram of elliptical Hohmann transfers.

1.4.3. Parabolic and Hyperbolic Hohmann Transfer

The same equations can also be used for transferring between an elliptical/circular orbit to a hyperbolic or parabolic orbit. Note that both of the orbits must lie on the same apse line as one another, examples of these types of orbit are shown in **Figure 3**.

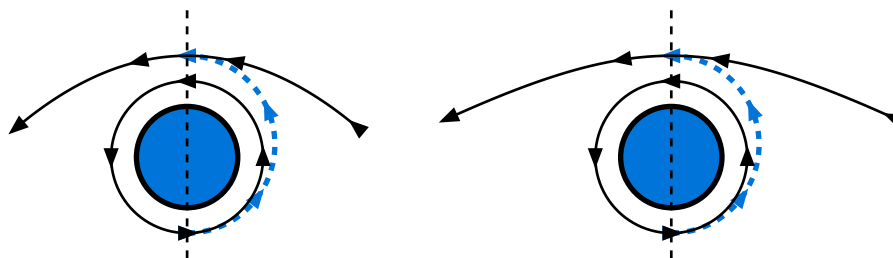


Figure 3: Schematic diagram of parabolic Hohmann transfer [Left] and hyperbolic Hohmann transfer [Right].

1.4.4. Bi-elliptic Hohmann Transfer

If the maximum number of burns is increased to three, then in some cases the **bi-elliptic Hohmann transfer** may be more efficient than a basic Hohmann transfer. This type of maneuver is shown in **Figure 4**

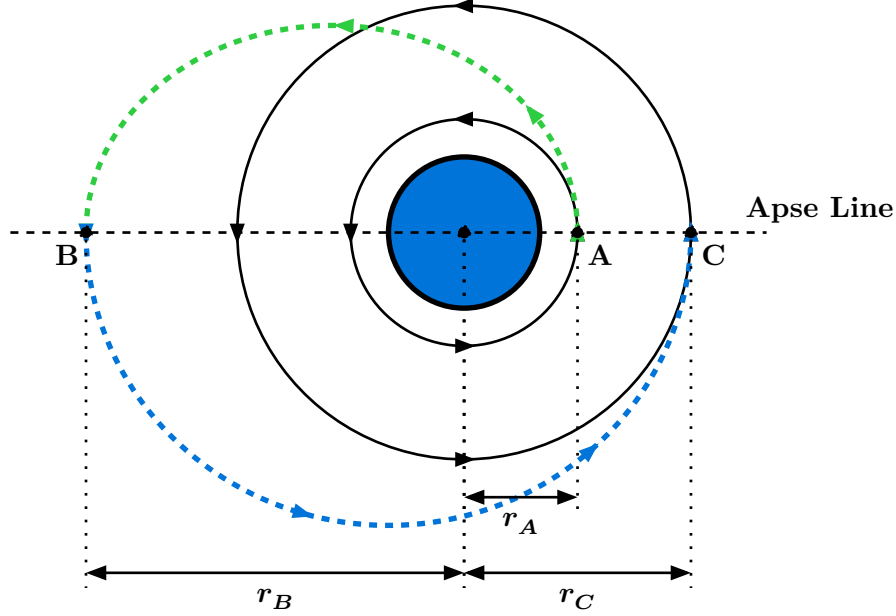


Figure 4: Schematic diagram of bi-elliptic Hohmann transfer.

As r_b goes to infinity, the Δv required at **B** goes to zero, allowing for a cheaper burn. However, a bi-elliptic Hohmann transfer is not always more efficient and so a derivation is required.

To start with, it is observed there are two transfer orbits T_1 from $A \rightarrow B$ and T_2 from $B \rightarrow C$, the perigee and apogee radii of these two transfer orbits is shown in **Eq. 7**.

$$T_1 \begin{cases} r_a = r_B \\ r_p = r_A \end{cases} \quad T_2 \begin{cases} r_a = r_C \\ r_p = r_B \end{cases} \quad (7)$$

From these definitions the change in velocity at point **A** (Δv_A) can be calculated and is shown in **Eq. 8** (Note, V_0 = velocity in origin orbit, V_1 = perigee velocity in T_1).

$$V_0 = \sqrt{\frac{\mu}{r_A}} \quad V_1 = \frac{h}{r_p} = \frac{1}{r_p} \sqrt{2\mu \frac{r_a r_p}{r_a + r_p}} = \sqrt{2\frac{\mu}{r_p^2} \frac{r_a r_p}{r_a + r_p}} = \sqrt{2\frac{\mu}{r_A} \frac{r_B}{r_A + r_B}} \quad (8.1)$$

$$\Delta v_A = V_1 - V_0 = \sqrt{2\frac{\mu}{r_A} \frac{r_B}{r_A + r_B}} - \sqrt{\frac{\mu}{r_A}} = \sqrt{\frac{\mu}{r_A}} \left[\sqrt{2\frac{r_B}{r_A + r_B}} - 1 \right] \quad (8.2)$$

$$\therefore \Delta v_A = V_0 \left[\sqrt{2\frac{r_B}{r_A + r_B}} - 1 \right] \quad \text{Where } V_0 = \sqrt{\frac{\mu}{r_A}} \quad (8.3)$$

The derivation for the change in velocity at point **B** (Δv_B) can be determined similarly, and is shown in **Eq. 9** (Note, V_1 = apogee velocity in T_1 , V_2 = apogee velocity in T_2).

$$V_1 = \frac{h}{r_a} = \sqrt{2 \frac{\mu}{r_B} \frac{r_A}{r_A + r_B}} \quad V_2 = \frac{h}{r_a} = \sqrt{2 \frac{\mu}{r_B} \frac{r_C}{r_C + r_B}} \quad (9.1)$$

$$\Delta v_B = V_2 - V_1 = \sqrt{2 \frac{\mu}{r_B} \frac{r_C}{r_C + r_B}} - \sqrt{2 \frac{\mu}{r_B} \frac{r_A}{r_A + r_B}} \quad (9.2)$$

$$\therefore \Delta v_B = \sqrt{2 \frac{\mu}{r_B}} \left[\sqrt{\frac{r_C}{r_C + r_B}} - \sqrt{\frac{r_A}{r_A + r_B}} \right] \quad (9.3)$$

The derivation for the change in velocity at point **C** (Δv_C) can be determined similarly, and is shown in **Eq. 10** (Note, V_2 = perigee velocity in T_2 , V_3 = velocity in destination orbit (further note that $V_C = V_2 - V_3$ to keep Δv +ve)).

$$V_2 = \frac{h}{r_p} = \sqrt{2 \frac{\mu}{r_C} \frac{r_B}{r_C + r_B}} \quad V_3 = \sqrt{2 \frac{\mu}{r_C}} \quad (10.1)$$

$$\Delta v_C = V_2 - V_3 = \sqrt{\frac{\mu}{r_C}} \left[\sqrt{2 \frac{r_B}{r_C + r_B}} - 1 \right] \quad (10.2)$$

The total Δv of this manuever would be the sum of **Eq. 8**, **Eq. 9** & **Eq. 10**, however some substitutions are made to make the resulting equations simpler, these substitutions are shown in **Eq. 11**.

$$\text{Let } \beta = \frac{r_B}{r_A} \quad \alpha = \frac{r_C}{r_A} \quad V_0 = \sqrt{\frac{\mu}{r_A}} \quad \therefore \quad (11.1)$$

$$\Delta v_A = V_0 \left[\sqrt{2 \frac{\beta}{\beta + 1}} - 1 \right] \quad (11.2)$$

$$\Delta v_B = V_0 \sqrt{\frac{2}{\beta}} \left[\sqrt{\frac{\alpha}{\beta + \alpha}} - \sqrt{\frac{1}{\beta + 1}} \right] \quad (11.3)$$

$$\Delta v_C = V_0 \sqrt{\frac{1}{\alpha}} \left[\sqrt{2 \frac{\beta}{\beta + \alpha}} - 1 \right] \quad (11.4)$$

Note that in **Eq. 11** a common term of V_0 can be extracted. Adding these expressions whilst multiplying V_B by $\sqrt{\frac{2}{\beta}}$ and V_C by $\sqrt{\frac{1}{\alpha}}$ yields **Eq. 12**.

$$\Delta v = \Delta v_A + \Delta v_B + \Delta v_C \quad (12.1)$$

$$= V_0 \left[\sqrt{\frac{2\beta}{\beta + 1}} - \underset{\substack{\text{Multiply by } \beta/\beta}}{\mathbf{1}}} + \sqrt{\frac{2\alpha}{\beta(\beta + \alpha)}} - \underset{\substack{\text{Multiply by } \alpha/\alpha}}{\mathbf{1}} + \sqrt{\frac{2}{\beta(\beta + 1)}} - \underset{\substack{\text{Combine Aqua}}{\mathbf{1}}} + \sqrt{\frac{2\beta}{\alpha(\beta + \alpha)}} - \underset{\substack{\text{Multiply by } \beta/\beta}}{\sqrt{\frac{1}{\alpha}}} \right] \quad (12.2)$$

$$\therefore \sqrt{\frac{2\alpha^2}{\alpha\beta(\beta + \alpha)}} + \sqrt{\frac{2\beta^2}{\alpha\beta(\beta + \alpha)}} = \sqrt{\frac{2(\alpha + \beta)}{\alpha\beta}} \quad -1 - \sqrt{\frac{1}{\alpha}} = -\frac{1 + \sqrt{\alpha}}{\sqrt{\alpha}} \quad (12.3)$$

$$\therefore \sqrt{\frac{2\beta^2}{\beta(\beta+1)}} - \sqrt{\frac{2}{\beta(\beta+1)}} = \sqrt{\frac{2}{\beta(\beta+1)}}(\beta-1) \quad (12.4)$$

Combining together the simplified expressions in **Eq. 12** yield the final expression for the Δv cost for a bi-elliptic Hohmann transfer shown in **Eq. 13**

$$\Delta v = V_0 \left[\sqrt{\frac{2(\alpha+\beta)}{\alpha\beta}} - \frac{1+\sqrt{\alpha}}{\sqrt{\alpha}} + \sqrt{\frac{2}{\beta(\beta+1)}}(\beta-1) \right] \quad (13.1)$$

$$\text{Where :} \quad V_0 = \sqrt{\frac{\mu}{r_A}} \quad \beta = \frac{r_B}{r_A} \quad \alpha = \frac{r_C}{r_A} \quad (13.2)$$

By plotting β against α , it is possible to see when the bi-elliptic Hohmann transfer is more efficient than a normal Hohmann transfer, this plot is shown in **Figure 5**.

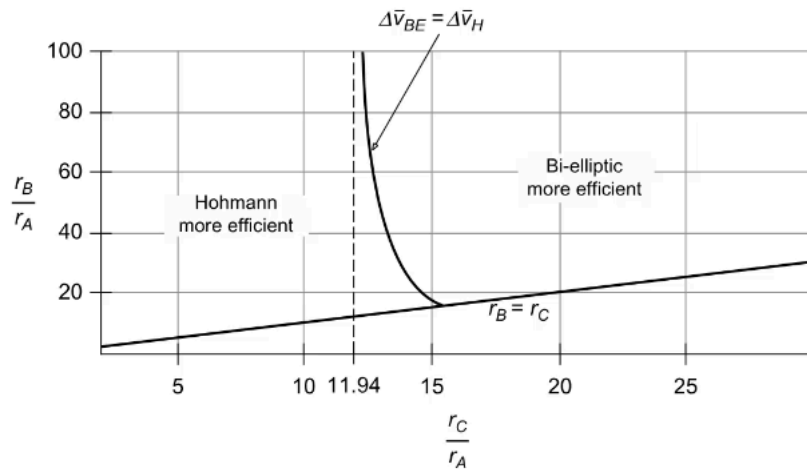


Figure 5: Plot of β against α for a bi-elliptic Hohmann transfer.

In **Figure 5** the straight line labelled $r_B = r_C$ is a normal Hohmann transfer. Specifically if $\alpha > 11.94$ then a **bi-elliptic Hohmann transfer is more efficient than a normal Hohmann transfer**.

1.5. Phasing Manuevers

The goal of a phasing maneuver is to **change the true anomaly** of a satellite **without changing the orbit** itself. This is useful for **rendezvous missions** or **repositioning** two satellites in the same orbit. The complexity of this problem is that changing the velocity of the orbit changes the orbit itself. The goal for such a maneuver is shown in **Figure 6**

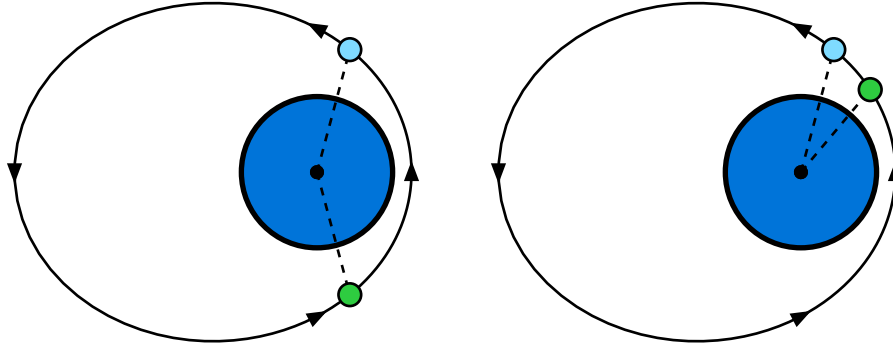


Figure 6: Schematic diagram showing the orbit of two satellites before [Left] and after [Right] a phasing maneuver.

One possible solution for this problem is using a **Lambert Solver** which will give a valid orbit between two points. However, this will often yield an orbit which will be expensive in terms of Δv .

Instead a **Hohmann Transfer** is used. If the orbit is raised relative to the original orbit then this introduces a lag and vice versa for lowering the orbit. Such a maneuver is shown in **Figure 7**.

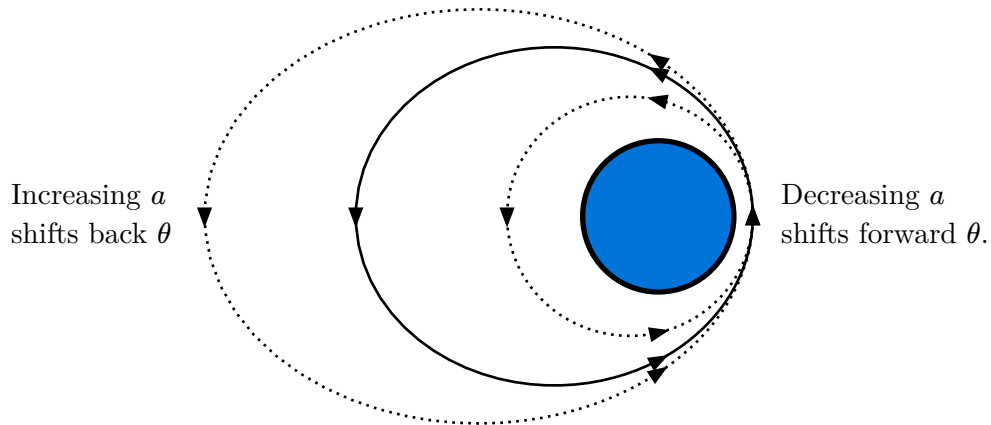


Figure 7: Schematic diagram showing the effect of increasing and decreasing the semi major axis.

To calculate the Δv of such a maneuver the following steps can be used:

1. Calculate time period T of phasing orbit:
 - Determine the change in time needed to get the object from its current θ to the desired theta. **Eq. 14** can be used.

$$\Delta t = T \pm t(\theta = \theta_0 \rightarrow \theta = \theta_1) \quad (14)$$

$\uparrow$
 + to increase θ
 - to decrease θ

- Set the time period of the phasing orbit (T_{Phasing}) as multiples of Δt depending on the required number of phasing orbits ($\uparrow n \rightarrow \downarrow \Delta v_{\text{Phasing}}$).

$$T_{\text{phasing}} = n\Delta t \quad (15)$$

2. Compute the semi major axis of the phasing orbit:
 - This can be done by using a modified form of the orbital period equation, shown in **Eq. 16**.

$$T = 2\pi\sqrt{\frac{a^3}{\mu}} \rightarrow a = \left(\frac{T\sqrt{\mu}}{2\pi}\right)^{\frac{3}{2}} \quad (16)$$

3. Compute the radius perigee or radius apogee using the semi major axis:
 - The phasing burn must take place at r_p or r_a of the original orbit so **Eq. 17** can be used to calculate the other orbital parameter of the phasing orbit.

$$2a = r_p + r_a \quad (17)$$

4. Use the Hohmann transfer equation to calculate $\Delta v_{\text{Phasing}}$ (Note that burn is symmetric so $\Delta v_{\text{Total}} = 2\Delta v_{\text{Phasing}}$).

1.6. Non-Hohmann Transfers

Whilst being **less efficient** Non-Hohmann transfers allow for more complex maneuvers such as moving between any two points on an orbit and moving between any two orbits. These maneuvers are more costly as the Δv vector is not only scaled but also rotated.

If two orbits intersect at least once then **only one burn** is needed to move between them. Non intersecting orbits require multiple burns as well as optimizations of starting points, end points and flight times.

1.6.1. Apse Line Rotation

The apse line is a straight line which connects together the apogee and perigee of an orbit. Rotating the Aps line of an orbit is an example of a non-Hohmann transfer. The derivation for the **required Δv** for this maneuver (including an optional shape change of the destination orbit) is shown below, starting with **Figure 8**.

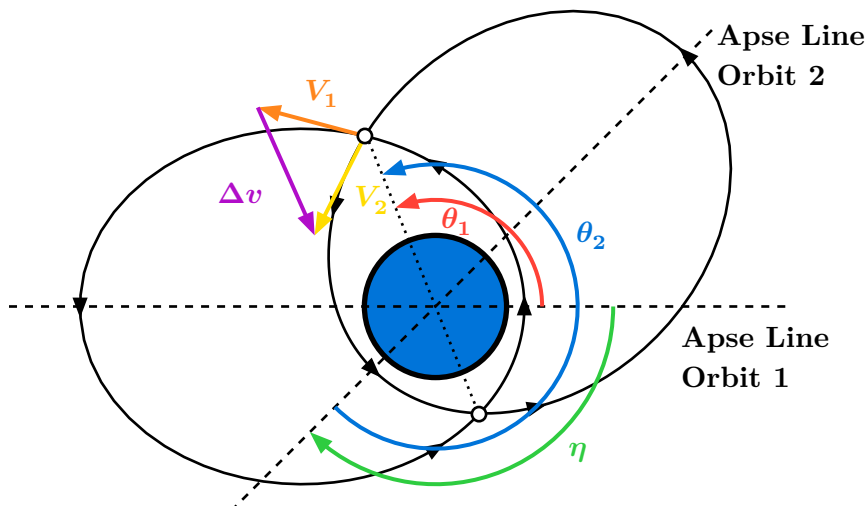


Figure 8: Schematic diagram showing an apse line rotation.

The first step in calculating the Δv of such a maneuver is to find the location of the intersection points, this is done in **Eq. 18**

$$\eta = \theta_1 - \theta_2 \quad \therefore \quad \theta_1 = \eta + \theta_2 \quad \theta_2 = \theta_1 - \eta \quad (18.1)$$

$$r_{\text{intersection}} = \frac{h_1^2 / \mu}{1 + e_1 \cos(\theta_1)} = \frac{h_2^2 / \mu}{1 + e_2 \cos(\theta_2)} \quad (18.2)$$

After rewriting θ_2 in terms of θ_1 and η , cross multiplying and simplifying, **Eq. 18** can be simplified into **Eq. 19**.

$$h_1^2 - h_2^2 = -h_1^2 e_2 \cos(\theta_1 - \eta) + h_2^2 e_1 \cos(\theta_1) \quad (19)$$

Eq. 19 can be further simplified by separating out $\cos(\theta_1 - \eta)$ using the angle difference equation and then rewriting the equation, grouping together terms in terms of θ_1 , this is shown in **Eq. 20**.

$$\cos(\theta_1 - \eta) = \cos(\theta_1) \cos(\eta) + \sin(\theta_1) \sin(\eta) \quad (20.1)$$

$$\therefore h_1^2 - h_2^2 = [e_1 h_2^2 - e_2 h_1^2 \cos(\eta)] \cos(\theta_1) + [-e_2 h_1^2 \sin(\eta)] \sin(\theta_1) \quad (20.2)$$

$$\therefore a \cos(\theta_1) + b \sin(\theta_1) = c \quad \begin{cases} a = e_1 h_2^2 - e_2 h_1^2 \cos(\eta) \\ b = -e_2 h_1^2 \sin(\eta) \\ c = h_1^2 - h_2^2 \end{cases} \quad (20.3)$$

The equation shown at the end of **Eq. 20** has a general solution for θ_1 , this is shown in **Eq. 21**

$$\theta_1 = \phi \pm \arccos\left(\frac{c}{a} \cos(\phi)\right) \quad \begin{cases} a = e_1 h_2^2 - e_2 h_1^2 \cos(\eta) \\ b = -e_2 h_1^2 \sin(\eta) \\ c = h_1^2 - h_2^2 \\ \phi = \arctan\left(\frac{b}{a}\right) \end{cases} \quad (21)$$

Note that the **Eq. 21** generates two solutions as there are two intersection points and the equations themselves are general. Now that a general expression for the intersection points of the two orbits has been determined, the velocity vector at this point in each orbit must be determined, this is done using the **flight path angle** (the angle between the velocity and the local horizon), shown in **Figure 9**.

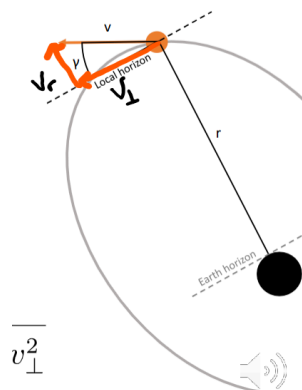


Figure 9: Schematic diagram showing how the flight path angle (γ), transverse velocity ($v_{\perp}$) and radial velocity (v_r).

The equations for the flight path angle, transverse velocity and radial velocity are shown below in **Eq. 22**.

$$\gamma = \arctan\left(\frac{v_r}{v_{\perp}}\right) \quad v_{\perp} = \frac{h}{r} \quad v_r = \frac{\mu}{h} e \sin(\theta) \quad v = \sqrt{v_r^2 + v_{\perp}^2} \quad (22)$$

The Δv of the maneuver can then be determined by looking at the triangle formed by the three velocity vectors, this section of the derivation is shown in **Eq. 23** and **Figure 10**.

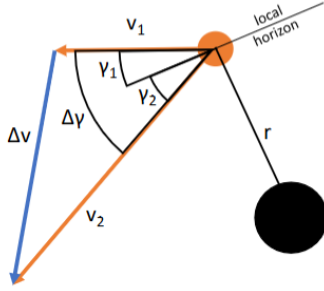


Figure 10: Schematic diagram showing how the velocity triangle formed by the Δv , v_1 and v_2 .

Note that the final equation in **Eq. 23** for the Δv is the same as the cosine rule which is illustrated further by **Figure 10**. The key equations for an Apse line rotation maneuver are shown in **Eq. 24**.

$$\theta_1 = \phi \pm \arccos\left(\frac{c}{a} \cos(\phi)\right) \quad \begin{cases} a = e_1 h_2^2 - e_2 h_1^2 \cos(\eta) \\ b = -e_2 h_1^2 \sin(\eta) \\ c = h_1^2 - h_2^2 \\ \phi = \arctan\left(\frac{b}{a}\right) \end{cases} \quad (24.1)$$

$$\gamma = \arctan\left(\frac{v_r}{v_{\perp}}\right) \quad \begin{cases} v_{\perp} = \frac{h}{r} \\ v_r = \frac{\mu}{h} e \sin(\theta) \\ v = \sqrt{v_r^2 + v_{\perp}^2} \end{cases} \quad \Delta v = \sqrt{v_2^2 + v_1^2 - 2v_1 v_2 \cos(\gamma_2 - \gamma_1)} \quad (24.2)$$

1.6.2. Special Case for the Apse Line Rotation

If the second orbit has an η angle of 0° , $\phi = 0$ and the equation reduce to the ones shown in **Eq. 25**.

$$\theta_1 = \pm \arccos\left(\frac{c}{a}\right) = \pm \arccos\left(\frac{h_1^2 - h_2^2}{e_1 h_2^2 - e_2 h_1^2}\right) \quad (25)$$

This maneuver is shown visually in **Figure 11** where the second orbit is different. Note the orange line would be an equivalent Hohmann maneuver. Despite the **Hohmann maneuver being more efficient**, there may be cases where this apse line maneuver is necessary (only one burn possible, restricted zones, time constraints).

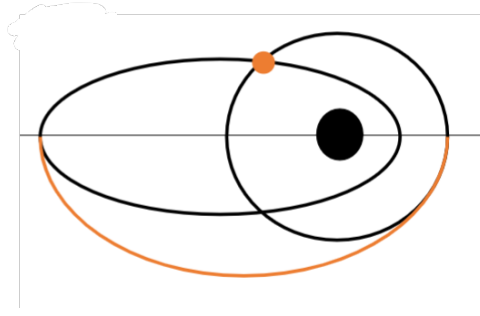
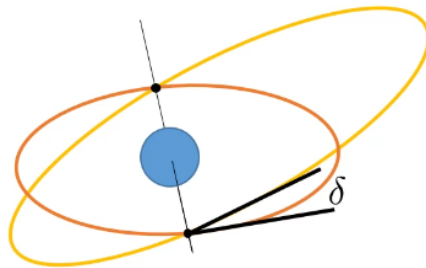


Figure 11: Schematic diagram showing an apse line maneuver where $\eta = 0^\circ$ and the orbits are different.

1.6.3. Plane Change Manoeuvre

A plane change manoeuvre is a three dimensional manoeuvre out of the plane of the orbit to alter the plane of the orbit. This follows a similar process to the phase change manoeuvre where the phase change manoeuvre, where the old and new velocity vectors are found and Δv is the difference between these two vectors. An angle δ is defined as the **plane change angle**, shown in **Figure 12** and in **Eq. 26**.



$$\cos(\delta) = \frac{\vec{h}_1 \cdot \vec{h}_2}{h_1 h_2} \quad (26)$$

Figure 12: Definition of plane change angle δ between an origin and destination orbit in a plane change manoeuvre.

1.6.3.1. Rigid Rotation of the Orbit

A simple plane change manoeuvre is one where the orbit is only rotated and the shape (apogee/perigee) about a rotation line which is a line in the original orbit plane that must intersect the orbit body. This means there are only ever two points within an orbit where a plane change manoeuvre can occur, this is shown in **Figure 13**.

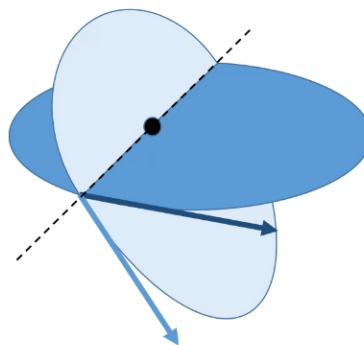
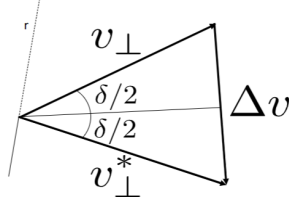


Figure 13: Rigid plane change with the rotation axis shown as the dotted line.

The velocity vector at the plane change manoeuvre point can be split into its radial and orthogonal components. The radial velocity vector is in the same direction as the rotation axis itself and so the rotation solely acts on the orthogonal component. The ΔV required for this manoeuvre is therefore given by the following equation **Eq. 27** and velocity triangle shown in **Figure 14**.



$$\Delta v = |\vec{v}^* - \vec{v}| = |\vec{v}_{\perp}^* - \vec{v}_{\perp}| \quad (27.1)$$

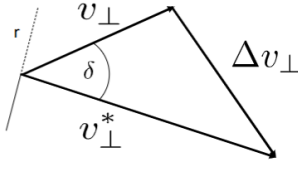
$$\therefore \Delta v = 2v_{\perp} \sin\left(\frac{\delta}{2}\right) \quad (27.2)$$

Figure 14: Velocity triangle for a rigid plane change manoeuvre.

Note that in this case the new orthogonal velocity ($\vec{v}_{\perp}^*$) and the old orthogonal velocity ($\vec{v}_{\perp}$) are the same as there is no shape change. Note that the Δv required for this manoeuvre is proportional to the velocity at that point in the orbit, yielding a very high value of Δv .

1.6.3.2. General Plane Change Manoeuvre with Shape Change

For a manoeuvre with a shape change, the cosine rule can be utilized as well as considering a possible change in the radial direction. The resulting equation and velocity triangle are shown in **Figure 15** and in **Eq. 28** respectively. Note that in **Eq. 28**, the second form is achieved by substituting the flight path angles into the first form.



$$\Delta v = \sqrt{(v_r^* - v_r)^2 + v_{\perp}^{*2} + v_{\perp}^2 - 2v_{\perp}^* v_{\perp} \cos(\delta)} \quad (28.1)$$

$$\Delta v = \sqrt{v^{*2} + v^2 - 2v^* v [\cos(\gamma^* - \gamma) - \cos(\gamma^*) \cos(\gamma) (1 - \cos(\delta))]} \quad (28.2)$$

Figure 15: Velocity triangle for a general plane change manoeuvre with a shape change.

Note that as this is effectively an equation for a general manoeuvre, applying certain conditions should reduce the equation to the known cases. Firstly if $v_r = v_r^* = 0$ and $\delta = 0$ the orbit plane rotation equation about a common apse line is generated.

1.6.3.3. Restricted Three Impulse Circular Plane Change

Plane change manoeuvres are expensive due to them being proportional to $v_{\perp}$, which is typically the largest component of v . Minimizing the $v_{\perp}$ can therefore minimize the Δv cost of the plane change manoeuvre, one way of doing this is doing the plane change manoeuvre at apogee. Thus arises the three impulse plane change manoeuvre, making use of a Hohmann transfer to raise the apogee, a plane change manoeuvre at apogee and then a burn at perigee to recircularize the orbit, a Schematic of this manoeuvre is shown in **Figure 16**.

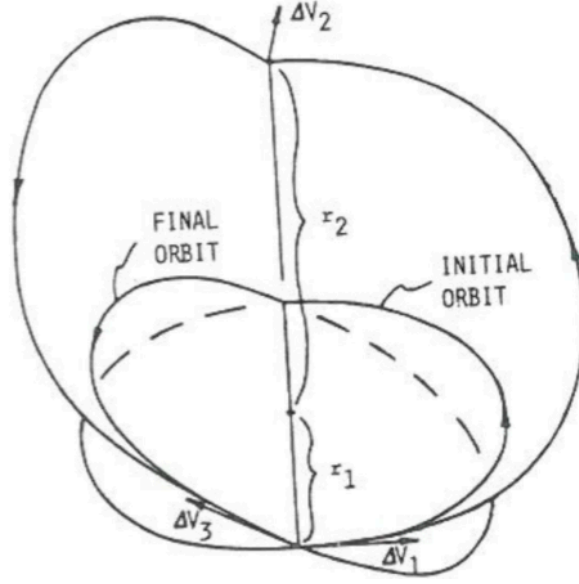


Figure 16: Diagram of a three impulse plane change manoeuvre.

To calculate the total Δv cost of this manoeuvre, the Vis-Viva equation can be used, shown in **Eq. 29**, rearranged for v .

$$\varepsilon = \frac{v^2}{2} - \frac{\mu}{r} = -\frac{\mu}{2a} \rightarrow v = \sqrt{\frac{2\mu}{r} - \frac{\mu}{a}} \quad (29)$$

This equation can then be applied to each point to work out the Δv . One important distinction in this manoeuvre is that Δv_1 is the same as Δv_3 as the plane change manoeuvre (Δv_2) doesn't change the orthogonal velocity just rotates it. Δv_1 and Δv_3 are thusly derived in **Eq. 30**, where v_c is the orbital velocity in the circular origin (and destination orbit).

$$\Delta v_1 = \Delta v_3 = v_1 - v_c = \sqrt{\frac{2\mu}{r_1} - \frac{\mu}{r_1 + r_2}} - \sqrt{\frac{\mu}{r_1}} = \sqrt{\frac{\mu}{r_1}} \left[\sqrt{\frac{2\frac{r_2}{r_1}}{1 + \frac{r_2}{r_1}}} - 1 \right] \quad (30)$$

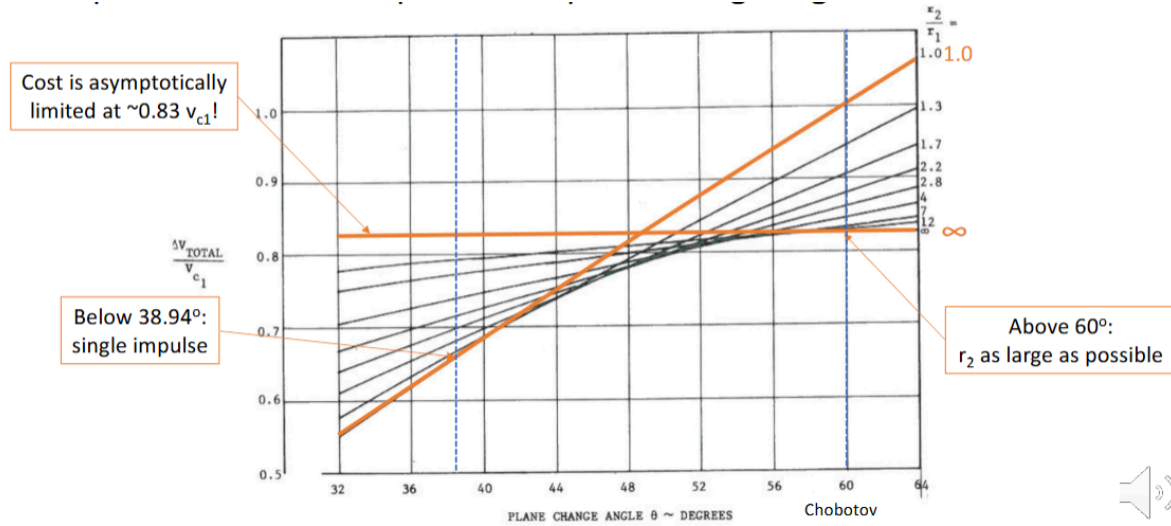
Δv_2 can be calculated by utilizing the rigid plane change equation as well as observing that at apogee the radial velocity is zero. This is shown in **Eq. 31**.

$$\Delta v_2 = 2v_{\text{Apogee}} \sin\left(\frac{\delta}{2}\right) = 2\sqrt{\frac{2\mu}{r_2} - \frac{\mu}{r_1 + r_2}} \sin\left(\frac{\delta}{2}\right) = 2\sqrt{\frac{\mu}{r_1}} \sqrt{\frac{2}{\frac{r_2}{r_1} \left(1 + \frac{r_2}{r_1}\right)}} \sin\left(\frac{\delta}{2}\right) \quad (31)$$

The total Δv is therefore the sum of two lots of **Eq. 30** and one lot of **Eq. 31**, the final result being shown in **Eq. 32**.

$$\Delta v = 2\sqrt{\frac{\mu}{r_1}} \left[\sqrt{\frac{2\rho}{1 + \rho}} - 1 + \sqrt{\frac{2}{\rho(1 + \rho)}} \sin\left(\frac{\delta}{2}\right) \right] \quad \text{Where } \rho = \frac{r_2}{r_1} \quad (32)$$

Plotting various values of ρ and δ yield the plot shown in **Figure 17**. This plot shows that the three impulse plane change manoeuvre is only more efficient than a single burn when $\delta > 38.94^\circ$ and that for $\delta > 60^\circ$, r_2 should be as large as possible.


 Figure 17: Plot of three impulse plane change manoeuvres for various ρ_s and δ_s

1.6.3.4. Hohmann Transfer with Split Change Manoeuvre

In a combination of both of these manoeuvres, it is not immediately clear for a combination of r_1 and r_2 where the plane change manoeuvre should be. In reality the plane change manoeuvre can be done at each burn by a certain angle. This manoeuvre is shown in **Figure 18**.

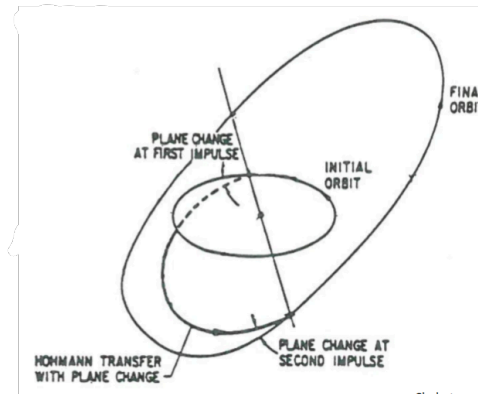


Figure 18: Schematic view of a split plane change manoeuvre combined with a Hohmann transfer.

The angular momentum and orbital velocities of each point of interest in each orbit are shown in **Eq. 33**.

$$h_0 = \sqrt{\mu r_1} \quad h_1 = \sqrt{2\mu \frac{r_1 r_2}{r_1 + r_2}} \quad h_2 = \sqrt{\mu r_2} \quad (33.1)$$

$$v_0 = \frac{h_0}{r_1} \quad v_{0,t} = \frac{h_1}{r_1} \quad v_{2,t} = \frac{h_1}{r_2} \quad v_2 = \frac{h_2}{r_2} \quad (33.2)$$

The proportion of the plane change that would be done at each manoeuvre is given by the α parameter. Using this the Δv for the entire manoeuvre can be determined, this is shown in **Eq. 34**.

$$\delta_0 = \alpha\delta \quad \delta_2 = (1 - \alpha)\delta \quad (34.1)$$

$$\Delta v_0 = \sqrt{v_0^2 + v_{0,t}^2 - 2v_0v_{0,t}\cos(\delta_0)} \quad \Delta v_2 = \sqrt{v_2^2 + v_{2,t}^2 - 2v_2v_{2,t}\cos(\delta_2)} \quad (34.2)$$

$$\Delta v = \Delta v_0 + \Delta v_2 = \sqrt{v_0^2 + v_{0,t}^2 - 2v_0v_{0,t}\cos(\delta_0)} + \sqrt{v_2^2 + v_{2,t}^2 - 2v_2v_{2,t}\cos(\delta_2)} \quad (34.3)$$

The value of Δv can be minimized by altering the value of α , the resulting minimization equation is shown in **Eq. 35**. Note that the equation itself would need some numerical analysis to solve for the α value that yields the minimum value of Δv .

$$\frac{d}{d\alpha}(\Delta v) = \frac{v_0v_{0,t}\delta\sin(\delta_0)}{\sqrt{v_0^2 + v_{0,t}^2 - 2v_0v_{0,t}\cos(\delta_0)}} - \frac{v_2v_{2,t}\delta\sin(\delta_2)}{\sqrt{v_2^2 + v_{2,t}^2 - 2v_2v_{2,t}\cos(\delta_2)}} = 0 \quad (35)$$

1.7. When and How to do Manoeuvres

Doing energy gain manoeuvres at perigee results in more energy gain, this is called the Oberth effect. This occurs as kinetic energy is proportional to the square of the velocity not the velocity on its own. An example derivation is shown in **Eq. 36**.

$$\Delta E = \frac{1}{2}m(v + \Delta v)^2 - \frac{1}{2}mv^2 = \frac{1}{2}m(v^2 + 2v\Delta v + \Delta v^2) - \frac{1}{2}mv^2 \quad (36.1)$$

$$= \frac{1}{2}m(v^2 + 2v\Delta v + \Delta v^2 - v^2) = \frac{1}{2}m(2v\Delta v + \Delta v^2) \quad (36.2)$$

The last equation in **Eq. 36** has v within it meaning a higher velocity results in a higher energy gain. Combining manoeuvres together in a single point is often advantageous for both the Oberth effect as well as for reducing vector addition losses. However this is not always the case, for example plane change manoeuvres are more efficient at high apogees as their Δv cost is proportional to the velocity.

2. Chapter 2: Orbit Perturbations and Propagations

2.1. Orbit Perturbations

For Keplerian orbits, the planet is assumed to be a point mass with no external forces acting. However for real spacecraft and satellites in orbit there exist disturbances which affect the motion of satellites in orbit. Though the central force from the body often dominates, some other perturbations include:

- Aerodynamic drag
- Solar radiation pressure
- Gravitational perturbations from a non-uniform, non-spherical central body
- Third body perturbations
- Electromagnetic drag
- Relativistic effects
- Yarkovsky-O'Keefe-Radzievskii-Paddack (YORP) effect from thermal radiation emission
- Outgassing, leaking and low thrust

A plot of the relative strength of these various disturbances for a LEO 2U CubeSat are shown in **Figure 19** and for a larger satellite in **Figure 20**.

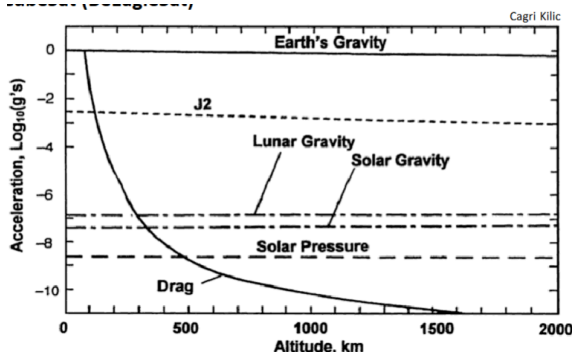


Figure 19: Relative strength of perturbations on a 2U CubeSat in LEO.

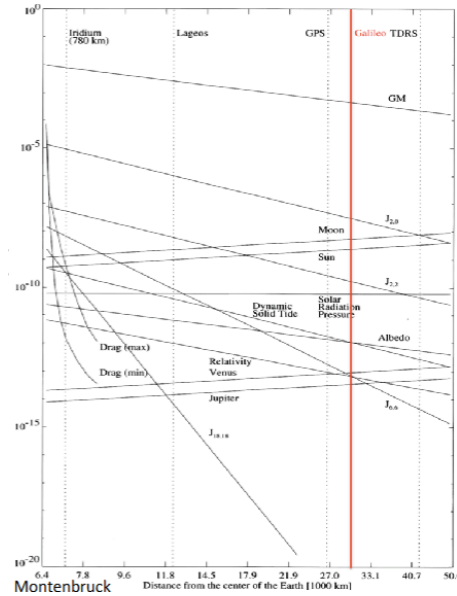


Figure 20: Relative strength of perturbations on a large satellite in LEO.

2.1.1. Aerodynamic Drag

The interaction of a spacecraft with the upper atmosphere can sap away energy from the orbit. Below 500km and even more significantly below 150km, drag acts to decrease the apogee of the orbit. The equation for drag is shown in **Eq. 37**.

$$a_{\text{drag}} = \frac{1}{2} \rho(r) v^2 C_d \frac{A}{m} \quad (37)$$

Typically $C_d \approx 2.2$ and the A/m is assumed to be an average parameter. The air density ρ itself is a parameter based on several variables such as position, altitude, solar activity etc. One simplified model of the air density is shown in **Eq. 38**.

$$\rho = \rho_0 \exp\left(-\frac{h}{H}\right) \quad H = \frac{kT}{mg} \quad (38)$$

Where H is the scale height. Another problem arises as $H(T)$ and $T(r)$. The temperature throughout the atmosphere varies as shown in **Figure 21**.

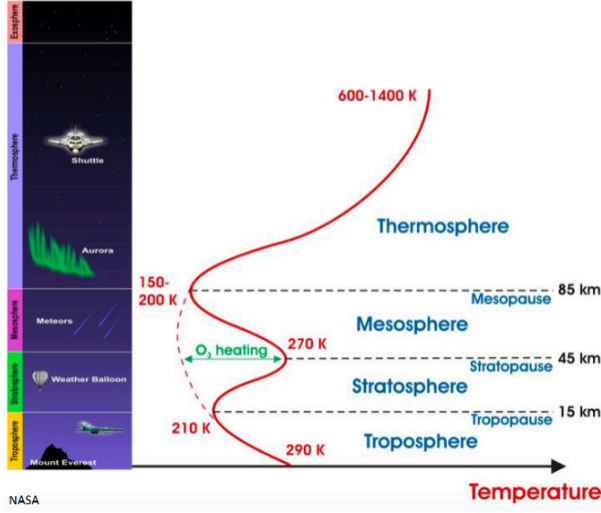


Figure 21: Plot of temperature throughout the atmosphere.

TABLE 8-4. Exponential Atmospheric Model. Although a very simple approach, this method yields moderate results for general studies. Source: Wertz, 1978, 820, which uses the *U.S. Standard Atmosphere* (1976) for 0 km, CIRA-72 for 25–500 km, and CIRA-72 with exospheric temperature, exospheric temperature, $T_{\infty} = 1000$ K for 500–1000 km. The scale heights have been adjusted to maintain a piecewise-continuous formulation of the density.

Altitude h_{avg} (km)	Base Altitude h_0 (km)	Nominal Density ρ_0 (kg/m ³)	Scale Height H (km)	Altitude h_{avg} (km)	Base Altitude h_0 (km)	Nominal Density ρ_0 (kg/m ³)	Scale Height H (km)
0–25	0	1.225	7.249	150–180	150	2.070×10^{-9}	22.523
25–30	25	3.899×10^{-2}	6.349	180–200	180	5.464×10^{-10}	29.740
30–40	30	1.774×10^{-2}	6.682	200–250	200	2.789×10^{-10}	37.105
40–50	40	3.972×10^{-3}	7.554	250–300	250	7.248×10^{-11}	45.546
50–60	50	1.057×10^{-3}	8.382	300–350	300	2.418×10^{-11}	53.628
60–70	60	3.206×10^{-4}	7.714	350–400	350	9.518×10^{-12}	53.298
70–80	70	8.770×10^{-5}	6.549	400–450	400	3.725×10^{-12}	58.515
80–90	80	1.905×10^{-5}	5.799	450–500	450	1.585×10^{-12}	60.828
90–100	90	3.396×10^{-6}	5.382	500–600	500	6.967×10^{-13}	63.822
100–110	100	5.297×10^{-7}	5.877	600–700	600	1.454×10^{-13}	71.835
110–120	110	9.661×10^{-8}	7.263	700–800	700	3.614×10^{-14}	88.667
120–130	120	2.438×10^{-8}	9.473	800–900	800	1.170×10^{-14}	124.64
130–140	130	8.484×10^{-9}	12.636	900–1000	900	5.245×10^{-15}	181.05
140–150	140	3.845×10^{-9}	16.149	1000–	1000	3.019×10^{-15}	268.00

Figure 22: Table of piecewise density model data.

A slightly better approach to model the atmosphere is a piecewise model whereby for a given altitude a base height is provided creating a piecewise density model, shown in **Figure 22** and in **Eq. 39**.

$$\rho = \rho_0 \exp\left(-\frac{h - h_0}{H}\right) \quad (39)$$

For high speed reentry of spacecraft the above models are insufficient as the atmosphere is not an ideal gas. For high speed upper atmosphere re-entry (Knudsen number ≥ 1) a Direct Simulation Monte-Carlo (DSMC) is used to extract temperatures dynamic pressures etc. Below this at lower speeds (Knudsen number < 1) CFD is used. Note the **main parameter to control the atmospheric drag is the $\frac{A}{m}$ ratio**.

2.1.2. Solar Radiation Pressure

Photons from the sun impart a small amount of momentum when they collide with a surface. Solar sails make use of this to gain Δv directly from sunlight instead of a thruster. The momentum change by a photon being absorbed as well as being reflected is shown in **Eq. 40**.

$$\Delta p_{\text{absorbed}} = \frac{E}{c} \quad \Delta p_{\text{reflected}} = \frac{2 \cos(\alpha) E}{c} \quad (40)$$

These equations can be written in terms of pressure by using $P = F/A$, $F = \Delta p / \Delta t$ and $A_{\text{eff}} = A \cos(\alpha)$ to yield **Eq. 41**. Note that for absorption the resulting force is in the direction of the light whereas for reflection the resulting force is normal to the reflective surface.

$$P_{\text{Absorption}} = \frac{W}{c \left(\frac{r}{1AU} \right)^2} \cos(\alpha) \quad P_{\text{Reflection}} = 2 \frac{W}{c \left(\frac{r}{1AU} \right)^2} \cos(\alpha)^2 \quad (41)$$

2.1.3. Gravitational Perturbations

2.1.3.1. Issued with Modelling as a Point Mass

If the planet is a sphere of uniform density, or a sphere with shells of varying density then both of these scenarios allow for modelling of said planet as a point mass. The force generated by this equivalent point mass is the same as the central body. However, there are further perturbations such as equator bulge, moving molten mantel, seas and mountains which all perturb the gravitational field.

2.1.3.2. Mesh Grid Modelling

If there exist high resolution images of the body, then a mesh grid approach can be used where finite elements mesh the body into small chunks. Equations can be solved over these chunks to generate gravitational fields. Though this is good for irregular bodies and does generate a high fidelity model, the generation of said model is very computationally intensive and also requires understanding of the internal structure of the body.

2.1.3.3. Surrogate Modelling

If instead the exact structure is not known of the body a surrogate model can be used which approximates the body as a collection of shapes with constant density. The gravitational potential field can then be determined easily for the body with a low computational cost at the downside of low fidelity.

2.1.3.4. Spherical Harmonics

A more complex method for modelling spherical bodies is spherical harmonics. Effectively a three dimensional Fourier transform where modes are added to approximate the gravitational behaviour of a spherical body. An example of some of these modes are shown in **Figure 23**.

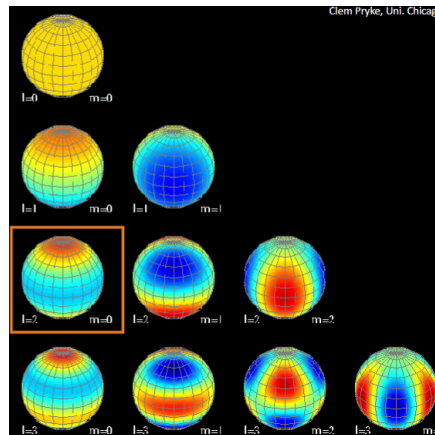


Figure 23: Different spherical harmonic modes, with J2 emphasized

The most significant of these perturbations is J2 which describes the bulging out of the equator of the planet due to its rotation. Its effect on the orbit is a net precession of the RAAN (right ascension of the ascending node (Ω) is the point where the orbit cuts the equator). This allows for sun synchronous orbits where the position of the sun relative to the spacecraft doesn't change.

2.1.4. Third Body Perturbations

Apart from the central body, the spacecraft also feels a gravitational force from third bodies. To calculate the magnitude and direction of these forces, an Ephemerides table is used which gives the location of a given planet at a certain time. The acceleration on the two main bodies is shown in **Eq. 42** and graphically in **Figure 24**.

$$\vec{a}_1 = Gm_2 \frac{\vec{R}_2 - \vec{R}_1}{|\vec{R}_2 - \vec{R}_1|^3} + Gm_3 \frac{\vec{R}_3 - \vec{R}_1}{|\vec{R}_3 - \vec{R}_1|^3} \quad (42.1)$$

$$\vec{a}_2 = Gm_1 \frac{\vec{R}_1 - \vec{R}_2}{|\vec{R}_1 - \vec{R}_2|^3} + Gm_3 \frac{\vec{R}_3 - \vec{R}_2}{|\vec{R}_3 - \vec{R}_2|^3} \quad (42.2)$$

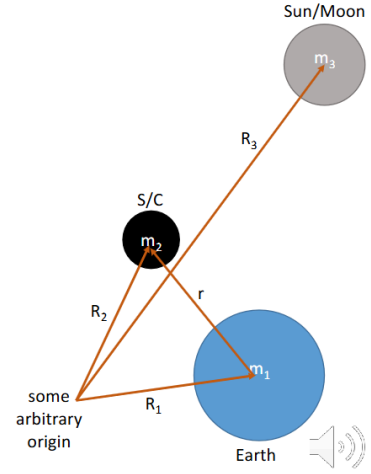


Figure 24: Schematic diagram of the three body problem.

The effect of the third body solely on the orbit of the spacecraft can be obtained by differentiating the difference in accelerations, shown in **Eq. 43**. This is when the spacecraft is set as the reference and the non-keplerian portion of the acceleration is singled out.

$$\vec{a} = \vec{a}_2 - \vec{a}_1 = \vec{a}_{\text{Keplerian}} + \vec{a}_{\text{third body}} \quad \vec{a}_{\text{third body}} = \mu_3 \left(\frac{\vec{r}_3 - \vec{r}}{|\vec{r}_3 - \vec{r}|^3} - \frac{\vec{r}}{|\vec{r}|^3} \right) \quad (43)$$

Note that in **Eq. 43**, the first term describes the acceleration of the spacecraft and the second the acceleration of the primary.

2.1.5. Specialized Orbit Perturbations

- **Electromagnetic Drag:** Conductor moving through a magnetic field sees eddy currents produced within it. This acts to create a drag like effect.
- **Special and General Relativity:** Travelling close to the speed of light changes motion from Newton. Travelling close to the speed of light changes the motion as well.
- **Yarkovsky Effect and YORP:** Anisotropic heat distribution of rotating body leads to effective radiation pressure causing a perturbation on asteroids. YORP causes spinning of asteroids due to a similar effect, over centuries may be the reason they break up.

2.2. Orbit Propagation

To implement the force models mentioned in the perturbation section, there needs to be a method to transform the force models into orbital motion. The three methods to do this are:

- **Analytical solutions:** Close formed equations that give exact motion.
- **Numerical methods:** Use an integrator to propagate motion
- **Averaged semi-numerical methods:** Use a mixture of both above methods to average out motion where possible allowing for larger numerical timesteps.

There also exists two types of perturbation theory, these are:

- **Special Perturbation Theory:** Numerically integrate a perturbation as a force. good for visualisation.
- **General Perturbation Theory:** Expand a series of perturbations as forces or potentials. Allow for broader analysis of inputs.

2.2.1. Special Perturbation

2.2.1.1. Direct Numerical Integration

In this methods the equation of motion with perturbation terms are directly appended onto the equations of motion, forming a 2nd order ODE in 3 dimensions (x,y,z) or a first order ODE in 6 dimensions (x,y,z and velocities). Though simple to implement, this scheme is slow and can drift over time.

2.2.1.2. Encke's Method

This method uses the known analytical solution and then uses the difference between the perturbed motion and the analytical solution to propagate motion. This method is better than direct numerical integration but is harder to implement, requiring normalisation after a while.

2.2.2. General Perturbation

Instead of the previous methods, this scheme utilizes a 6D phase space where small perturbations yield small changes in the orbit. The Osculating elements allow for movement from one orbit to another due to a perturbation via a tangential elliptical orbit at that point. These osculating elements are shown in **Figure 25** and **Figure 26**.

$$\begin{aligned} \frac{da}{dt} &= 0 \\ \frac{de}{dt} &= 0 \\ \frac{di}{dt} &= 0 \\ \frac{d\Omega}{dt} &= 0 \\ \frac{d\omega}{dt} &= 0 \\ \frac{dM}{dt} &= n \end{aligned}$$

Figure 25: Kaplerian element rates.

$$\begin{aligned} \frac{da}{dt} &= \frac{2e \sin \theta}{n\sqrt{1-e^2}} a_r + \frac{2a\sqrt{1-e^2}}{nr} a_s \\ \frac{de}{dt} &= \frac{\sqrt{1-e^2} \sin \theta}{na} a_r + \frac{\sqrt{1-e^2}}{na^2 e} \left(\frac{a^2(1-e^2)}{r} - r \right) a_s \\ \frac{di}{dt} &= \frac{r \cos(\theta + \omega)}{na^2 \sqrt{1-e^2}} a_w \\ \frac{d\Omega}{dt} &= \frac{r \sin(\theta + \omega)}{na^2 \sqrt{1-e^2} \sin i} a_w \\ \frac{d\omega}{dt} &= -\frac{\sqrt{1-e^2} \cos \theta}{nae} a_r + \frac{a(1-e^2)}{eh} \left[\sin \theta \left(1 + \frac{1}{1+e \cos \theta} \right) \right] a_s - \frac{r \cot i \sin(\theta + \omega)}{na^2 \sqrt{1-e^2}} a_w \\ \frac{dM}{dt} &= n - \frac{1}{na} \left(\frac{2r}{a} - \frac{1-e^2}{e} \cos \theta \right) a_r - \frac{(1-e^2) \sin \theta}{nae} \left(1 + \frac{r}{a(1-e^2)} \right) a_s \end{aligned}$$

Figure 26: Equations of motion of osculating elements.

Note that the perturbing accelerations are in the spacecraft plane coordinate system. Instead of Gauss's formula, Lagrange provides a similar expression only limited to potentials. These equations can be solved analytically but are typically very large, so instead numerical methods or a hybrid averaging method are used.